



90s Kid

If you are a 90s kid and played a lot of violent video games, then worry you are not radicalised says this new study. <http://dgit.in/kifof90s>



Battle of Emojis

How difficult can it be for a new Emoji to make its mark out there? Not too difficult? Think again. <http://dgit.in/EmojiBtl>

Military tech

pendence on missiles and instead pursued improved manoeuvrability by incorporating high-tech fly-by-wire systems, powerful flight computers, and powerful thrust vectoring engines that allowed jet fighters of that era to deliver significantly better dogfighting capability. This was followed by an evolutionary upgrade to this generation dubbed 4.5 generation that further enhanced fighter manoeuvrability by incorporating inherently canard-equipped unstable delta wing designs that require powerful avionic systems and flight control computers just to keep these aircraft from falling out of the skies. All these enhancements enabled these jets to achieve insane angle of attacks, rapid climb rates, and ultimate manoeuvrability.

The mid-generation upgrade also added superior Active Electronically Scanned Array (AESA) radars and stealthy, yet precise Infra-Red Search and Track (IRST) systems to hunt enemy aircraft more effectively. The use of exotic lightweight, yet high strength engineering materials enabled engineers to not only increase the overall performance, but also subject the aircraft to greater g-forces, thereby allowing for even tighter and previously impossible levels of agility. This generation also saw the use of radar absorbing paints and materials that significantly reduced the radar signature and made these fighters difficult to track and detect with conventional radar systems.

Stealth capable jet fighters

Cut to the present and fifth-generation fighters essentially combine all the defining characteristics of 4.5 generation aircraft such as advanced avionics, high-tech airframes built with exotic high-strength materials, inherently unstable and flight computer-dependent design for the control surfaces, undetectable inter-aircraft communication systems and such with a more focused approach to stealth. First-generation stealth technology has existed before fifth-generation fighters were introduced, but this was restricted mainly to dedicated bombers and strictly ground attack jets such as the B-2 Spirit bomber and F-117 attack jet. These aircraft achieved stealth with a host of radar absorbing materials and airframe design geared towards deflecting radar waves away from the source.

However, these design considerations limited manoeuvrability and speed, which prevented the application of true stealth technology in jet fighters, which are meant to be agile and fast. This is where fifth-generation fighter approach to stealth comes in, which covers all aspects of flying undetected but also involves moving quickly and nimbly while being able to track enemy aircraft without opening themselves for detection. This includes advanced Low Probability of Intercept Radar (LPIR) systems that enable an offensive stealth approach that isn't just limited to ground targets. Another important aspect of the new generation fighters involves the ability to achieve higher situational awareness by the means of battlefield networking technology that allows these jets to communicate without blowing their cover.

Why is stealth needed?

Now that we know what entails a fifth-generation air superiority jet fighter, it's worthwhile to find out exactly why Japan is pushing to develop its own frontline stealth fighter. Let's not forget that we're dealing with a nation that shies from the very notion of war. One that goes to euphemistic extremes by referring to its armed forces in a painfully benign fashion as Japan Ground Self-Defense Force (JGSDF), where the recruits are largely trained with airsoft replica guns instead of the real deal. It's hard to swallow the fact that the same country that depends upon US military for protection would want to produce something as advanced as a fifth-generation stealth fighter.

One hypothesis points towards the rising hostilities and threat from North Korea prompting some Japanese politicians eager to break Japan's age old non-proliferation policy and develop nuclear weapons. Possessing a viable stealth air superiority platform is a great asset under such circumstances. Another catalyst for

the decision comes from the fact that the F-22 manufacturer Lockheed-Martin was prohibited from selling the air superiority fighter to Japan. Interestingly, unconfirmed reports peg the F-22 maker to be involved in the development of the Shin-Shin X-2 technology demonstrator.

Another possible reason could be the current US military's rapidly diminishing power and edge, as evidenced by the recent



The ShinShin will hopefully be the defender of Japanese skies

annexure of Crimea, Russian foray into the Middle East, and being surpassed by the Chinese economy and challenged by its growing military might. To put this into perspective, the US air force would have needed 150 aircraft to protect Taiwan against Chinese air force back in 1996. Today, the estimated number stands at nearly 2000, which makes it nigh impossible for the USAF to realistically deploy enough aircraft to rescue Taiwan from a Chinese aerial onslaught. It makes perfect sense for the Japanese to invest in means to reduce its dependence on the US for security.

Let's not forget that the US government will be selling the fifth-generation F-35 Lightning II to the Japanese, but the platform has gone seriously over-budget and is marred with performance issues and damning glitches. Not to mention, the fact that it also famously got outclassed by the F-16, a jet fighter built in the '80s. The overpriced and highly underperforming F-35 purchase, therefore, is only a stop-gap solution until the Japanese induct their own fifth-generation air superiority fighter.

Crouching tiger, radar invisible dragon

However, the most pressing compulsion behind the Japanese rush to procure a